JCUIScaler 1.0 – Risk Analysis (updated November 2025, RC1.2, RC2 and 1.0 updates November 2026)

1. Introduction

This document presents the current risk analysis for JCUIScaler 1.0, updated on the basis of the current Technical Specification and the present state of the code after refactoring. These risks are key to preserving deterministic behaviour, stability and architectural consistency.

2. Critical Architecture Risks

Note (2026-11-08): the rules in force for contributors, derived from this section and its updates, are maintained in the CONTRIBUTING.md file in the project repository.

- Changing the model from object to class (breaks the architecture, life cycle, heap).

- Violating the deterministic recursion of ScaleControl.

- Changing the PageControl or grid algorithms.

- Introducing constructs that depend on DPI or on LCL auto-scaling.

- Changing the OnShow/OnResize mechanics in a way that affects DPI stabilisation.

3. Functional Risks Concerning Components

- ListView – column instability across widgetsets.

- TreeView – item height differences depending on the widgetset.

- StatusBar – inconsistent panel widths.

- GroupBox – different Caption offsets depending on the platform.

- BorderSpacing / Padding / Margins – potential conflicts with Anchors.

- Splitter – unstable behaviour at extreme form sizes.

- Notebook – different implementations across widgetsets.

4. Technical Risks

- Overloading ScaleControl with too many conditions.

- Excessive recursion cost for forms with thousands of controls.

- Incorrect handling of minimum form sizes (360×300).

- Violating the no-heap-allocation rule.

- Overly aggressive optimisations that change behaviour.

5. Regression Risks Compared with JCUIScaler Beta

- Loss of deterministic scaling.

- Changed PageControl behaviour (especially tsButtons).

- Unstable grid column proportions.

- Errors after Maximize → Restore.

6. Environmental and Testing Risks

Current tests are performed on Windows 11 with the standard LCL widgetset. This is the reference environment; however, differences may appear on Linux and macOS, especially in the PageControl, GroupBox, TreeView and StatusBar controls.

- Widgetset differences may require additional scaling rules.

- Undetected DPI errors in other environments despite the code having no DPI dependencies.

- Differences in window auto-scaling behaviour between platforms.

7. Mitigation and Recommendations

- Regression tests against the Beta version

- Comparing behaviour across platforms.

- No modification of the core algorithms.

- Minimal use of RTTI.

- Performance analysis for very large forms.

8. Summary

The most important element of JCUIScaler 1.0 RC1 stability is preserving the deterministic scaling model and the absence of DPI dependencies. The critical risks concentrate on preserving the architecture (object, no heap), the stability of PageControl/grids and consistent behaviour

across environments. In its present form the document is fully prepared for work to begin in the remaining project threads.

Risk Analysis – Update to the RC1 Code (integrated)

2. Critical Architecture Risks – RC1 update

Based on the Beta code, all existing architecture risks were confirmed as still valid. Their description was extended with elements resulting from tightening the recursion, stabilising PageControl and grids, and the event algorithms. All risks remain critical.

- Violating the deterministic recursion of ScaleControl (strengthened in RC1).

- Introducing heap allocation or destructors (the object model must remain unchanged).

- Changing the PageControl or grid algorithms — key deterministic modules.

- Introducing DPI dependencies (Alpha 7 keeps full independence).

- OnShow / OnResize modifications that break sequence stabilisation.

3. Functional Risks – RC1 update

The RC1 code confirms that the earlier component risks are still valid. PageControl and grid stability was extended, but for the remaining components the risks are unchanged. New analyses confirm, among other things, high sensitivity to widgetset differences and to platform-dependent behaviour.

- ListView — item and column heights differ depending on the widgetset.

- TreeView — node heights not fully stable.

- StatusBar — panel behaviour differs between Windows and Linux.

- GroupBox — platform sensitivity when computing the Caption.

- Components with BorderSpacing/Anchors — the risk of conflicts was reduced in Alpha 7 but not eliminated.

- Splitter — edge-case behaviour with large window disproportions.

4. Technical Risks – Alpha 7 update

Alpha 7 made the algorithms more robust, but the technical risks remain valid, in particular for large control trees and recursion performance.

- Recursion overload with thousands of controls.

- Incorrect handling of minimum form sizes.

- Introducing expensive helper operations that reduce determinism.

- Violations of the no-heap-allocation rule.

- Changes in font scaling that disturb hierarchical propagation.

5. Regression Risks Compared with JCUIScaler Beta

Version RC1 complies with the principles of deterministic operation, but regression risks remain an important area of assessment. The PageControl and grid improvements reduce the risk but do not eliminate it.

- Loss of deterministic scaling on unusual forms.

- Changes in grid column proportions after many resizes (Alpha 7 reduces this risk but does not eliminate it entirely).

- Unexpected effects after Maximize → Restore.

- Errors scaling PageControl in complex tab configurations.

6. Environmental and Testing Risks – RC1 update

Widgetset differences remain a significant risk. Alpha 7 introduces no changes that break the multi-platform architecture, but some controls behave differently across platforms.

- PageControl not fully consistent between Windows, Linux and macOS.

- Different system font heights affecting grids and ListView.

- Automatic widgetset behaviour cannot be anticipated by JCUIScaler (in line with the architectural constraints).

7. Risk Mitigation Methods – RC1 update

The recommendations remain in line with the original version but are extended with the analysis of complex structures. RC1 improves determinism, which reduces some of the architecture risks.

- Comparative tests across platforms.

- Regression tests for PageControl and grids.

- Profiling the performance of recursive scaling.

- Full compliance with the architectural design principles (object, no heap).

Risk Analysis – RC1.2 Update (2026-11-07)

This update covers RC1.2 (source-level Delphi compatibility, BorderSpacing/GTK2/LocalRound fixes) and the first runtime tests outside Windows. Related documents: Technical Specification R1-05, Static Analysis Rev H, RoadMap Rev E.

2. Critical Architecture Risks – RC1.2 update

The critical risks remain fully valid. RC1.2 violates none of them: the object architecture, the ScaleControl recursion, the PageControl/grid algorithms and the OnShow/OnResize sequence are unchanged. The Delphi changes are limited to neutral code (an access class for Font, explicit variable initialisation) and three {$IFDEF FPC} blocks.

Incident of 2026-11-07 (recorded as a warning): a fix for the TListView-on-a-hidden-tab problem was proposed and implemented in a working version RC1.3. It made the scaler register an OnResize handler on every TListView control. The fix violated two critical risks at once — it changed the event mechanics (the scaler reacted to control events outside its three explicit calls) and tied the lifetime of the TJCUIScaler object to the controls (the end of the “object on the stack” model). The fix was rejected entirely and withdrawn; the problem was resolved with a configuration recommendation without code changes.

Conclusion and rule in force from RC1.2:

- Every proposed code change is verified against section 2 of this document before implementation.

- The library operates only within its three explicit calls (ScalerInit, ScalerOnShow, ScalerOnResize) and the public emergency method RestoreInitialScale; it registers no event handlers on controls or on the form.

- The lifetime of TJCUIScaler must not depend on the form's controls.

- Problems caused by LCL or widgetset behaviour are resolved first by a configuration recommendation or a description in the documentation, not by changing the core.

- The integration layer (component TJCScaler, R2-02) may hook form events, but only in order to call the three core methods; the core remains unchanged.

3. Functional Risks – RC1.2 update

Runtime tests on GTK2 confirmed some of the component risks and allowed them to be described precisely:

- ListView — risk materialised (Static Analysis F-18). A native TListView on a hidden tab sheet reports a ClientWidth without its vertical scrollbar; columns scaled at that time are wider by the scrollbar width until the next form resize (Windows and GTK2). Mitigation: ScrollBars = ssVertical (Specification, section 13.1). Residual risk: the end of the last column is temporarily clipped. On Windows, ScalerOnResize can optionally be called in TPageControl.OnChange; on GTK2 this call has no effect.

- Grids — the risk does not occur for hidden tab sheets: LCL grids manage their own scrollbars (measured for ssAutoBoth, ssVertical, ssBoth; 6 and 60 rows).

- TreeView — the compilation error for LCLGTK2 was fixed (the ItemHeight property does not exist in the LCL; correctly DefaultItemHeight, F-14). Node height scaling is still inactive (FS3) — risk unchanged.

- StatusBar — panels covered by the demo; panel scaling correct on GTK2. The risk of Windows/Linux differences is reduced, not eliminated.

- BorderSpacing / Anchors — two BorderSpacing bugs fixed (F-12: position stored instead of BorderSpacing; F-13: section unreachable for some types). New risk: Anchors = [] at a DPI different from DesignTimePPI makes the LCL move controls before the model is built (F-19). Mitigation: default Anchors = [akLeft, akTop] (Specification, section 13.1).

- GTK2 — ScrollBars = ssBoth means scrollbars are always visible (GTK_POLICY_ALWAYS); widgetset behaviour, documented (F-20).

- GroupBox, Splitter, Notebook — no new findings; Splitter and Notebook are not covered by the demo.

4. Technical Risks – RC1.2 update

- Minimum form size: in the code MIN_FORM_WIDTH = MIN_FORM_HEIGHT = 200 (the value 360×300 from the original version of this document is outdated).

- Rounding: LocalRound = Floor(V + 0.5) — incorrect rounding of negative values fixed (F-15); results for positive values unchanged.

- Compilation: before RC1.2 the library did not compile for LCLGTK2 — an error invisible when compiling only for Windows. Mitigation: test compilation with all supported widgetsets.

- FPC 3.3.1 (trunk): the object-type language patterns used in the library verified — compiles without warnings, no leaks. Risk of object being marked deprecated in the future: low.

5. Regression Risks – RC1.2 update

- Determinism: on GTK2 at 96 and 120 DPI there is no drift after resize cycles (pixel-exact screenshot comparison) — the risk of losing determinism did not materialise.

- RestoreInitialScale: public method (design decision — emergency mechanism). On GTK2, resetting the second form gives a state identical to its first show. To be verified on Windows: the order of OnResize relative to FForm.SetBounds inside RestoreInitialScale (hypothesis F-22 — with a synchronous WM_SIZE the proportional loop could be applied after the recomputation from the model).

- Maximize → Restore: unchanged, limitation documented.

6. Environmental and Testing Risks – RC1.2 update

The test environment was extended with Linux/GTK2 (FPC 3.2.2, LCL 3.0, virtual X server, 96 and 120 DPI). Windows 11 (Lazarus 4.8) remains the reference environment.

- Delphi VCL (unofficial): Delphi 13 compilation to be confirmed. Limitations: JCSCALER_BORDER_TAG ignored, no monitor-change handling, demo for Lazarus only.

- macOS (Cocoa) and Qt: not tested — risk unchanged, to be described in the README as untested.

- Monitor change (R2-01): moved beyond 1.0 — risk accepted and documented as a limitation.

7. Risk Mitigation Methods – RC1.2 update

- Every code change is verified against section 2 before implementation.

- Test compilation with all supported widgetsets (including LCLGTK2).

- Runtime tests outside Windows; pixel-exact screenshot comparison after resize cycles.

- Configuration recommendations in the documentation instead of core changes (Specification, section 13.1).

- A demo project covering all types with dedicated handling (including StatusBar panels and JCSCALER_BORDER_TAG).

Risk Analysis – RC2 Update (2026-11-07)

This update covers RC2: the TJCScaler component (integration layer), the Lazarus packages and the license files. Related documents: Technical Specification R1-06, Static Analysis Rev I, RoadMap Rev F.

2. Critical Architecture Risks – review of the TJCScaler component

The component was reviewed before and after implementation against section 2 and the rule from the RC1.2 update (Static Analysis F-23). Result: compliant.

- Object → class model: unchanged. TJCUIScaler remains an object type; the component (a class) holds it as a field — the core does not move to the class model.

- ScaleControl recursion, PageControl/grid algorithms: unchanged — the core code has not changed since RC1.2 (changes in comments only).

- OnShow/OnResize sequence: the component calls the core methods in the same order as the manual integration (the application handler first, then the scaler). Results pixel-identical to the manual integration (12/12 screenshots, GTK2).

- Events: the component hooks only form events; it registers no handlers on controls. The core reacts to no events other than its explicit calls.

- Lifetime: the TJCUIScaler object is a field of the component owned by the form; it does not depend on the controls.

- Compatibility: the core name TJCUIScaler is unchanged (production applications using the manual integration); the component is named TJCScaler.

3. Functional Risks – component

- OnShow chaining (F-24): the LCL has no handler list for OnShow. Risk: an application replacing OnShow at run time disconnects the scaler. Mitigation: the Attach method, documentation (Specification R1-06, sections 10 and 12.2). The AddHandlerFirstShow variant was rejected — it would change the moment the reference model is built.

- BeforeResize/AfterResize events: risk of misuse (changing the geometry of model controls in BeforeResize is overwritten). Mitigation: description in the documentation; AfterResize designated as the place for adjustments.

- One component per form, owner TForm — otherwise the component remains inactive (no run-time error).

5. Regression Risks – RC2 update

- RestoreInitialScale (F-22): the hypothesis of a reversed OnResize order on Windows was not confirmed — closed by the author's test; the code comment corrected.

- License headers: changes in comments only; the code of all files is identical to the previous version when comments are ignored.

6. Environmental, Licensing and Testing Risks – RC2 update

- Component installation in the IDE confirmed on Windows (Lazarus 4.8); the packages compile for GTK2 (Lazarus 3.0).

- License: identical to the LCL (GNU LGPL version 2 with the static linking exception). The earlier entry “LGPL 2.1” in the RoadMap was wrong — corrected (Static Analysis F-27). Risk of licensing inconsistency with Lazarus removed.

- Attribution: the copyright notice and the original author's name in the headers of all source files, with a prohibition on removing them; the notice travels with the files also with manual integration (copying only the .pas/.inc files).

- Delphi 13: compilation of the core still unconfirmed — planned for the end of the work. The component is available for the LCL only.

7. Risk Mitigation Methods – RC2 update

- Every change to the component is verified by a pixel-exact screenshot comparison with the manual integration.

- Functional extensions are implemented in the integration layer (events, virtual methods), not in the core.

- The core is frozen until the 1.0 release; only changes to comments and version labels are allowed.

Risk Analysis – 1.0 Update (2026-11-07)

This update covers the 1.0 release state and the verification with Delphi 13.2. Related documents: Technical Specification R1-08, Static Analysis Rev K, RoadMap Rev H.

2. Critical Architecture Risks – 1.0 update

- The Delphi compatibility fix (F-28) was checked against section 2: the change concerns only the type in an is test; the algorithm, the event sequence and the object lifetime are unchanged; behaviour under the LCL is identical (A/B test: 12/12 screenshots).

6. Environmental and Testing Risks – 1.0 update

- Delphi 13.2 (Win64): compilation and basic operation confirmed, including third-party controls (TMS UI Pack PageControl, RxDBGrid) — the risk of the code not compiling in Delphi is removed.

- Delphi/VCL: TSplitter descends from TGraphicControl and receives only the generic scaling — limitation documented (Specification R1-08, section 12.3).

- Line endings: source files in CRLF, enforced by .gitattributes — the risk of RAD Studio warnings and inconsistencies after cloning the repository is removed (F-29).

- macOS (Cocoa) and Qt: still untested — described in the README.

7. Risk Mitigation Methods – 1.0 update

- Every Delphi compatibility fix is verified by an A/B test under the LCL (pixel-exact screenshot comparison).

Risk Analysis – Document Closed for Version 1.0 (2026-11-08)

This update closes the document for version 1.0. The document is not updated on an ongoing basis; the current document is the Technical Specification (R1-09), and the rules for contributing changes are maintained in the CONTRIBUTING.md file.

2. Critical Architecture Risks – rules after the 1.0 release

- Clarification of the RC1.2 rule: the scaler operates only within explicit calls made by the application or the integration layer and registers no event handlers by itself. Adding a new explicit call (e.g. for a monitor change) is allowed.

- The core freeze applied to the 1.0 publication. After 1.0, changes to the core are allowed within section 2, with an A/B test (screenshot comparison) and a review by the project maintainer.

- R2-01 (monitor change) touches the OnShow/OnResize sequence, a critical area. Acceptance condition: without a monitor change the behaviour is identical to version 1.0 (A/B test).

- Font scaling (experimental): disabled by default; when disabled the behaviour is identical to version 1.0; determinism also applies in the experimental mode; behaviour described for every tested platform.

6. Environmental and Testing Risks – closing 1.0

- macOS 27 (Lazarus 4.8, Cocoa): demo verified — the risk of no testing on macOS removed. Tested platforms: Windows, macOS and Linux/GTK2; Delphi 13.2 unofficially. Qt — untested.

- CI (GitHub Actions): compilation of the packages, the library with LCLGTK2 and the demo on every change in the repository — protection against regressions like F-14 (failure to compile on a widgetset other than the one used by the author).